Dipolo elettrico

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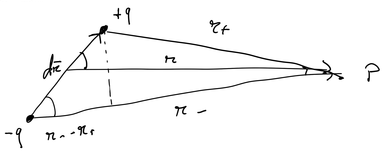


$p := q\,dr$ momento di DIPOLO

$$V(P) = \frac{1}{4\pi\varepsilon_0}\left(\frac{q}{r^+} - \frac{q}{r^-}\right) = \frac{q}{4\pi\varepsilon_0}\left(\frac{r^- - r^+}{r^+ r^-}\right)$$

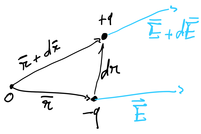
Se $r \gg dr$

$$V(P) = \frac{1}{4\pi\varepsilon_0}\,\frac{q\,dr\cos\theta}{r^2}$$

$$V(P) = \frac{1}{4\pi\varepsilon_0}\,\frac{\vec{p}\cdot\vec{r}}{r^3}$$

$$\vec{E} = -\nabla V = \frac{1}{4\pi\varepsilon_0}\left(\frac{3\vec{r}\cdot\vec{p}}{r^5}\vec{r} - \frac{\vec{p}}{r^3}\right)$$

INTERAZIONE DIPOLO - CAMPO



$$d\vec{r} = dx\,\vec{u}_x + dy\,\vec{u}_y + dz\,\vec{u}_z$$

$$d\vec{E} = dE_x\,\vec{u}_x + dE_y\,\vec{u}_y + dE_z\,\vec{u}_z$$

$$d\vec{E} = \vec{\nabla}\vec{E}\cdot d\vec{r} \rightarrow \begin{cases} dE_x = \vec{\nabla}E_x\cdot d\vec{r} \\ dE_y = \vec{\nabla}E_y\cdot d\vec{r} \\ dE_z = \vec{\nabla}E_z\cdot d\vec{r} \end{cases}$$

FORZA AGENTE SUL DIPOLO

$$\vec{F} = \vec{F}_- + \vec{F}_+ = -q\vec{E} + q(\vec{E}+d\vec{E})$$

$$= q\,d\vec{E} = q\,\vec{\nabla}\vec{E}\cdot d\vec{r} = \vec{p}\cdot\vec{\nabla}\vec{E} = \begin{cases} F_x = \vec{p}\cdot\vec{\nabla}E_x \\ F_y = \vec{p}\cdot\vec{\nabla}E_y \\ F_z = \vec{p}\cdot\vec{\nabla}E_z \end{cases}$$

Se E uniforme $\begin{cases} \nabla E_x = 0 \\ \nabla E_y = 0 \\ \nabla E_z = 0 \end{cases}$

$$\Rightarrow \vec{F} = 0$$

MOMENTO

$$\vec{\tau} = \vec{r}\times\vec{F}_- + (\vec{r}+d\vec{r})\times\vec{F}_+ = -r\times q\vec{E} + (r+dr)\times q(\vec{E}+d\vec{E})$$

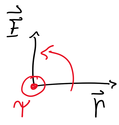
$$= \cancel{-q\vec{r}\times\vec{E}} + \cancel{q\vec{r}\times\vec{E}} + q\,\vec{r}\times d\vec{E} + q\,d\vec{r}\times\vec{E} + \cancel{q\,d\vec{r}\times d\vec{E}}$$ infinitesimo di ordine superiore

$F = q\,d\vec{E}$

$$\tau = \vec{r}\times\vec{F} + \vec{p}\times\vec{E}$$

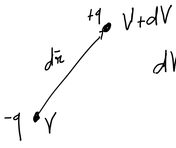
Se $\vec{E}$ uniforme

$$\vec{\tau} = \vec{p}\times\vec{E}$$



τ allinea p col campo.

Energia



$$dV = \nabla V\cdot d\vec{r}$$

Energia del dipolo

$$U = U_+ + U_- = q(V+dV) - qV = q\,dV = q\,\nabla V\cdot d\vec{r}$$

$$\nabla V = -E$$

$$U = -\vec{p}\cdot\vec{E}$$

Energia

$U_{min} = -\vec{p}\cdot\vec{E} = -pE\cdot\underset{=1}{\cos\alpha}$ $\alpha = 0$ $\uparrow p \uparrow E$

equilibrio stabile

$U_{max} = -\vec{p}\cdot\vec{E} = -pE\,\underset{=-1}{\cos\alpha} = pE$ $\alpha = \pi$ $\downarrow p \uparrow E$

equilibrio instabile